



Chapter 1 Real numbers

≡ Tags

Courses

Analysis 1.pdf

▼ Side math Resumes

Properties – 12–_:

1. $(p \wedge q) \iff (q \wedge p)$ ✓
2. $(p \vee q) \iff (q \vee p)$ ✓
3. $(p \wedge q) \wedge r \iff p \wedge (q \wedge r)$ ✓
4. $(p \vee q) \vee r \iff p \vee (q \vee r)$ ✓
5. $p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$ ✓
6. $p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$ ✓
7. $\neg(\neg p) \iff p$ ✓
8. $\neg(p \wedge q) \iff (\neg p) \vee (\neg q)$ ✓
9. $\neg(p \vee q) \iff (\neg p) \wedge (\neg q)$ ✓
10. $(p \implies q) \iff (\neg q \implies \neg p)$
11. $((p \implies q) \wedge (q \implies r)) \implies (p \implies r)$
12. $((p \iff q) \wedge (q \iff r)) \implies (p \iff r)$

✓ Proof

8* And 9* : *Morgan – Laws*

$p \equiv T$ Is called : Tautology

$p \equiv F$ Is called : Contradiction

The Law of Contradiction :

- $p \wedge \neg p \equiv F$

The Law of Excluded Middle :

- $p \vee \neg p \equiv T$

De Morgan's Laws :

- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
 - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
-

Double negation:

- $\neg(\neg p) \equiv p.$

Idempotent:

- $p \wedge p \equiv p, p \vee p \equiv p.$

Absorption 1:

- $p \vee (p \wedge q) \equiv p.$

Absorption 2:

- $p \wedge (p \vee q) \equiv p$
-

Order Of The 3's :

- $(p \vee q) \vee r \equiv p \vee (q \vee r).$
- $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r),$

Distribution :

- $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r),$

- $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

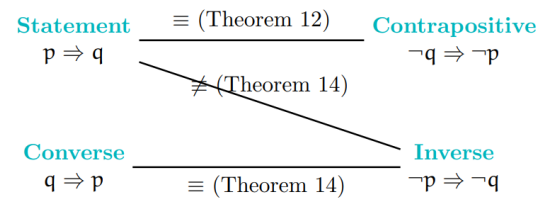
p	q	r	$p \wedge q$	$q \wedge r$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Converse :

- The converse of the statement $p \Rightarrow q$ is $q \Rightarrow p$.

Inverse :

- The inverse of the statement $p \Rightarrow q$ is $\neg p \Rightarrow \neg q$.



Contrapositive :

- The contrapositive of the statement $p \Rightarrow q$ is $\neg q \Rightarrow \neg p$.

Summary :

Logic Rules

1. $p \wedge \neg p \equiv F$
2. $p \vee \neg p \equiv T$
3. $\neg(\neg p) \equiv p$
4. $p \wedge q \equiv q \wedge p$
5. $p \vee q \equiv q \vee p$
6. $\neg(p \wedge q) \equiv \neg p \vee \neg q$
7. $\neg(p \vee q) \equiv \neg p \wedge \neg q$
8. $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
9. $(p \vee q) \vee r \equiv p \vee (q \vee r)$
10. $p \wedge p \equiv p, p \vee p \equiv p$
11. $p \vee (p \wedge q) \equiv p$
12. $p \wedge (p \vee q) \equiv p$
13. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
14. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
15. $p \oplus q \equiv q \oplus p$

Some Important Negations :

Statement	Negation
$\wedge$ (And)	$\vee$ (Or)
$A \Rightarrow B$ (Implicite)	$A \wedge \neg B$
$A \Leftrightarrow B$ (Equivalence)	$A \oplus B / (A \vee B) \wedge \neg(A \wedge B)$



notes

- عندما تكون العملية تبديلية، يكفي حساب طرف واحد فقط للتحقق.
- يمكن إثبات أنها حقل (زمرة أو حقل) $E \subseteq (F, *, D)$
- مجموعة - قواعد - بديهيات - نظريات
- $a < b \equiv aRb : R$ هي علاقة ترتيب
- prove by exhaustion (you exhaust every possibility) it means prove case by case
- prove by comprehension its means

▼ 1. Algebraic and order axiom

first session : 25/09/2024

The real number system consists of real numbers together with two elementary operations: addition (denoted by $+$) and multiplication (denoted by $\times$ or $\cdot$), and the less than relation (denoted by $<$).

- The system requires two basic elements: 0 and 1.
- If a and b are two real numbers, then $a + b$ and $a \times b$ are also real numbers.
- We say that R is closed (stable) under addition and multiplication.
- For any real numbers a and b , the statement $a < b$ is either true or false.
- We can also define subtraction and division in terms of $(+)$ and $(\times)$. Similarly, we can define $\geq$ and $>$ using $<$.

second session : 30/9/2024

▼ Remark 1

We have three categories of properties for R : algebraic properties, logic, and completeness. Let's start with the algebraic properties for $a, b, c \in R$:

1. $(a + b) + c = (b + c) + a$ (associative axiom for addition)
2. $a + 0 = 0 + a = a$ (identity axiom for addition)
3. For $a \in R$, there exists $-a \in R$ such that $a + (-a) = (-a) + a = 0$ (inverse axiom for addition)
4. $a \times (b \times c) = (a \times b) \times c$ (associative axiom for multiplication)
5. $a + b = b + a$ (commutative axiom for addition)
6. $a \times 1 = 1 \times a = a$ (identity axiom for multiplication)
7. For any $a \in R$, where $a \neq 0$, there exists $a^{-1} \in R$ such that $a \times a^{-1} = a^{-1} \times a = 1$ (inverse axiom for multiplication)
8. $a \times (b + c) = a \times b + a \times c$ (distributive axiom of multiplication over addition)
9. $a \times b = b \times a$ (commutative axiom for multiplication)
10. One and exactly one of the following holds: $a < b$, $a = b$, or $a > b$ (trichotomy axiom)
11. If $a < b$ and $b < c$, then $a < c$ (transitive axiom)

12. If $a < b$, then $a + c < b + c$ (addition and order axiom)

13. If $a < b$ and $c > 0$, then $ac < bc$ (multiplication and order axiom)

▼ Remark 2

$a=b$, means a is the same as b (logic).

axiom 1,2,3,4 allow $\mathbb{R}$ to get a commutative additive group structure

Axiom 5,6,7 allow $\mathbb{R}^*$ to get multiplicative group structure

Axiom 5,6,7,9 allow $\mathbb{R}^*$ to get a commutative group structure

Axiom 1,...,8 make $\mathbb{R}$ field



Theorem:

If a, b, c, d are real numbers with $c \neq 0$ and $d \neq 0$, then:

1. $ac = bc$ implies $a = b$
2. $a \cdot 0 = 0$
3. $-(-a) = a$
4. $(c^{-1})^{-1} = c$
5. $-(a) = -a$
6. $a(-b) = (-a)b = -ab$
7. $(-a) + (-b) = -(a + b)$
8. $(-a)(-b) = ab$
9. $(a/c) \times (b/d) = ab/cd$
10. $a/c + b/d = (ad + bc)/cd$

▼ 2. $\mathbb{R}$ is a totally ordered field

$\mathbb{R}$ is a totally ordered field. All the order properties of inequalities stem from the axioms defined below.

We can define $\geq$, $\leq$, and $>$ in terms of $<$:

If $a < b$, this means that $b > a$.

If $a \leq b$, this means $(a < b \text{ or } a = b)$.

$a \geq b$ means $(a > b \text{ or } a = b)$.



Theorem 1.2. If a , b and c are real numbers then

1. $a < b$ and $c < 0$ implies $ac > bc$
2. $0 < 1$ and $-1 < 0$
3. $a > 0$ implies $1/a > 0$
4. $0 < a < b$ implies $0 < 1/b < 1/a$
5. $|a + b| \leq |a| + |b|$ (triangle inequality)
6. $||a| - |b|| \leq |a - b|$ (a consequence of the triangle inequality)

2.1 Order relation

We defined in $\mathbb{R}$ an order relation $\leq$ by $a \leq b$ or $b \geq a$. We recall the axioms:

(A1) : $\forall a \in \mathbb{R} : a \leq a$

(Reflexive).

(A2) : $a \leq b$ and $a \geq b \iff a = b$ (Antisymmetric).

(A3) : $a \leq b$ and $b \leq c \iff a \leq c$

(transitive).

Example:

One can show the utility of such order relation

1. Let a be a real number such that $|a| < \varepsilon$, $\forall \varepsilon > 0$, we have $a = 0$ (elsewhere, if we choose $\varepsilon = |a|/2$ contradiction).
2. Let a and b be two real numbers such that $a < b + \varepsilon$, $\forall \varepsilon > 0$, then $a \leq b$ (else, if we take $\varepsilon = (b - a)/2$ we get a contradiction).

▼ 3. Natural Numbers and Induction

Mathematical induction is a mathematical proof technique requiring essentially that a statement $P(n)$ holds for every natural number $n = 0, 1, 2, 3, \dots$; that is, the overall statement is a sequence of infinitely many cases $P(0), P(1), P(2), P(3), \dots$

For any $n \geq 0$ we verify that $P(n)$ is true $\rightarrow P(n+1)$ true

Example:

$$1. \ 1+2+\dots+n=n(n+1)/2$$

first we prove $n=1$ $p(n)=1$ true

we suppose that $P(n)$ is true and prove $P(n+1)$ true

$$1+2+\dots+n+n+1=(n+1)(n+1)/2$$

then this true

▼ 4. Absolute value

The absolute value (or modulus) of any real number a , denoted by $|a|$, is defined as

$$a \longrightarrow |a| \in \mathbb{R}^+ = \begin{cases} a, & \text{si } a \geq 0 \\ -a, & \text{si } a < 0 \end{cases},$$

$$|a| = \sup (a, -a).$$

Fourth session: 7/10/2024



Theorem

For $a, b \in \mathbb{R}$, the following properties hold:

1. $|a| = 0 \Leftrightarrow a = 0$
2. $|a| = |-a|$
3. $|ab| = |a| \cdot |b|$
4. For $a > 0$, we have: $|x| \leq a \Leftrightarrow -a \leq x \leq a$
 - a. If $x \geq 0$, then $|x| = x$, and $x \leq a, x \geq -a$ ($-a \leq x \leq a$)
 - b. If $x \leq 0$, then $|x| = -x \leq a \Leftrightarrow -a \leq x$ ($-a \leq x \leq a$)
5. $|a + b| \leq |a| + |b|$, for all $a, b \in \mathbb{R}$

▼ Proof:

1. If a and b have the same sign and are positive: $a > 0, b > 0$, then $|a + b| = a + b, |a| = a, |b| = b$
2. If a and b have the same sign and are negative: $a < 0, b < 0$, then $|a + b| = -(a + b), |a| = -a, |b| = -b$
3. Let's suppose that $a < 0 < b$

$$-(a + b) \leq a + b \leq |a| + |b|$$

$$|a + b| \leq |a| + |b| \text{ (triangle inequality)}$$

Remark: One can easily prove the following generalized case of the triangle inequality for $a_i, i = 1, n$. We can show, by induction, that

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|.$$

▼ 5. Interval

In mathematics, a (real) interval is a set that contains all real numbers lying between any two numbers. More precisely:

Definition 1.2. Let a and b be two real numbers such that $b > a$.

1. The set $\{x : a < x < b\}$ is called an open interval and is denoted by (a, b) .
2. The set $[a, b] = \{x : a \leq x \leq b\}$ is called a closed interval (compact interval).

3. The sets $[a, b) = \{x : a \leq x < b\}$ and $(a, b] = \{x : a < x \leq b\}$ are called half-open intervals (right and left, respectively).

For all intervals, the points a and b are called endpoints.

If $a = b$, we define $[a, a] = \{a\}$ (degenerate closed interval) and $(a, a) = \emptyset$. The length of the interval (closed, open, or half-open) is given by the real number $b - a$.

Examples:

1. The set $\{x : x \leq a\}$ is a left-unbounded closed interval, denoted by $(-\infty, a]$.
2. The set $\{x : x < a\}$ is a left-unbounded open interval, denoted by $(-\infty, a)$.
3. The set $\{x : x \geq a\}$ is a right-unbounded closed interval, denoted by $[a, +\infty)$.
4. The set $\{x : x > a\}$ is a right-unbounded open interval, denoted by $(a, +\infty)$.
5. The set $\mathbb{R}$ is also denoted by $(-\infty, +\infty)$. Here, $-\infty$ and $+\infty$ represent infinity.

▼ 6. Archimedean property

This property does not follow from the algebraic and order axioms alone. It states, informally, that there are no real numbers beyond all natural numbers.

6.2 Archimedean axiom

For every real number a , there exists a natural number n such that $a < n$. Equivalently, the set $\mathbb{N}$ is not bounded above. We say that $\mathbb{R}$ is Archimedean.

Corollary

For all real numbers a and b such that $a > 0$, there exists $n \in \mathbb{N}$ such that $na > b$.

▼ Proof

Apply the axiom with $a = b/a$.

Let $x \in \mathbb{R}$. Then there exists a unique integer number, called the floor function,

$$E(x) = [x] = \lfloor x \rfloor$$

such that:

$$E(x) \leq x < E(x) + 1$$

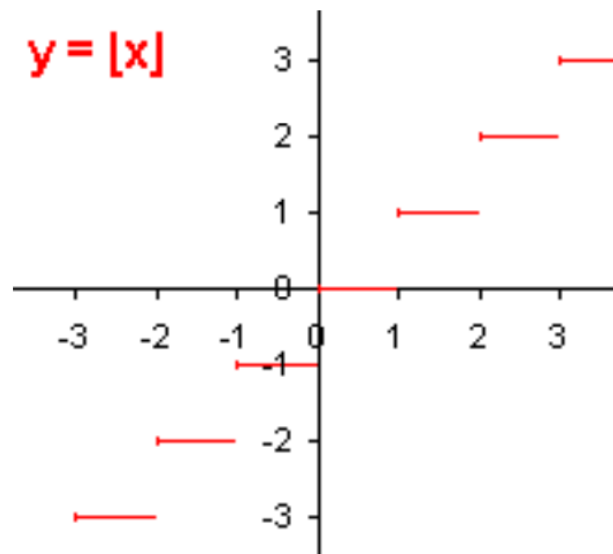


Figure 1.2: Graph of the floor function $Y = [X]$

Example 1.3. $E(e) = 2$, $E(-e) = -3$, $E(1.45632) = 1$

▼ 7. The completeness axiom

Fifth session: 08/10/2024

Definition: Let S be a subset of $\mathbb{R}$.

- We say that S is upper bounded if there exists a real number M such that for all x in S : $x \leq M$.
- We say that S is lower bounded if there exists a real number m such that for all x in S : $x \geq m$.
- The least element of the set of upper bounds is called the supremum of S and is denoted by $\sup S$.
- The greatest element of the set of lower bounds is called the infimum of S and is denoted by $\inf S$.
- If $\sup S$ belongs to S , it is called the maximum and is denoted by $\max S$ ($\max S = \sup S$).

- If $\inf S$ belongs to S , it is called the minimum and is denoted by $\min S$ ($\min S = \inf S$).

Example 1: $N = S = \{1, \dots\}$

Not lower bounded

Upper bounded by $(-\infty, 1]$

$\inf S = \min S = 1$

$\sup S$ does not exist

Example 2: $A = \{1/(n+2), n \geq 0\}$

Upper bounded by $[0.5, +\infty)$

Lower bounded by $(-\infty, 0]$

$\max A = \sup A = 0.5$

$\inf A = 0$, $\min A$ doesn't exist

Definition: A subset of $\mathbb{R}$ is called bounded if one of the following is satisfied:

1. $\exists m, M \in \mathbb{R} : m \leq x \leq M$, for all $x \in A$
2. $\exists c > 0, (c \in \mathbb{R}) : |x| \leq c$, for all $x \in A$

Example:

1. S is bounded because there exist $m = -3$ and $M = 2$ such that:

$-3 \leq x \leq 2$ for all $x \in S$

2. $\exists C : |x| \leq C$, for all $x \in S$

Let A be an upper bounded set. Then the supremum exists (every upper bounded set admits a supremum).

Corollary

Every lower bounded subset admits an infimum.



Theorem

Let A and B be two nonempty subsets of $\mathbb{R}$ such that $A \subseteq B$. We have:

1. $\sup A \leq \sup B$
2. $\inf A \geq \inf B$

▼ Proof:

For all x in A , $x \in B$, then $x \leq \sup B$.

Since $\sup B$ is an upper bound of A , and

$\sup A$ is the least upper bound of A , then $\sup A \leq \sup B$.

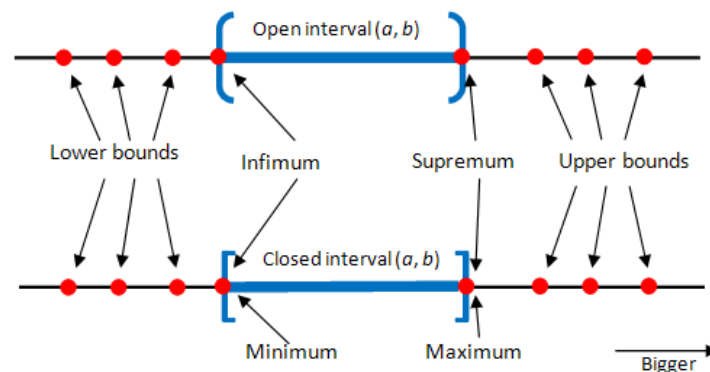
For all $x \in A$, $x \geq \inf B$. $\inf B$ is a lower bound of A , and since $\inf A$ is the greatest lower bound, then $\inf A \geq \inf B$.



Theorem

Let A and B be two bounded subsets of $\mathbb{R}$. Then we have:

1. $\sup(A \cup B) = \max \{\sup A, \sup B\}$.
2. $\inf(A \cup B) = \min \{\inf A, \inf B\}$.
3. If $A \cap B \neq \emptyset$, show that $\sup(A \cap B) \leq \min \{\sup A, \sup B\}$
4. $\inf(A \cap B) \geq \max \{\inf A, \inf B\}$.



6th session: 14/10/2024

Exercise 1.1. Let A and B be nonempty subsets of real numbers such that $A \subset B$, prove that

1. If B is upper bounded, the $\sup B$ exists and $\sup A \leq \sup B$.
2. If B is lower bounded, the $\inf B$ exists and $\inf B \leq \inf A$.

Exercise 1.2. Let A and B be nonempty subsets of real numbers, show that

1. $\sup(A \cup B) = \max \{ \sup A, \sup B \}$.
2. $\inf(A \cup B) = \min \{ \inf A, \inf B \}$.
3. If $A \cap B \neq \emptyset$, show that $\sup(A \cap B) \leq \min \{ \sup A, \sup B \}$ and $\inf(A \cap B) \geq \max \{ \inf A, \inf B \}$.

▼ 8. Characterization of Supremum and Infimum



Theorem

Let A be an upper-bounded subset of $\mathbb{R}$. The real number M is the supremum of A if the following hold:

1. For all $x \in A$, $x \leq M$
2. For any $\varepsilon > 0$, there exists $x \in A$ such that $M - \varepsilon \leq x \leq M$

Example: $A = \{n/(n+1), n \geq 1\}$

We show that $\sup A = 1$
 $n \geq 1 \rightarrow n + 1 \geq 2 \rightarrow 1/(n+1) \leq 1/2 \rightarrow 1 - 1/(n+1) \geq 1/2$,
 so for all $x = n/(n+1) \in A$: $x \geq 1/2$

We have for all $x = n/(n+1) \in A$, $n/(n+1) \leq 1$ = Let $\varepsilon > 0$. There exists $x = n_0/(n_0+1) \in A$ such that $1 - \varepsilon < n/(n+1) \dots \rightarrow \varepsilon > 1/(n+1) \rightarrow n > 1/\varepsilon - 1$ If $\varepsilon \geq 1$, (true for all n) If $0 < \varepsilon < 1$, by the Archimedean axiom of $\mathbb{R}$, n exists



Theorem

Let A be a lower-bounded subset of $\mathbb{R}$. The real number m is said to be the infimum of A if the following hold:

1. For all $x \in A$, $x \geq m$
2. For all $\varepsilon > 0$, there exists $x_\varepsilon \in A$ such that $m \leq x_\varepsilon \leq m + \varepsilon$

$$A = \{n/(n+1), n \geq 1\}$$

$$\inf A = \min A = 1/2$$

$$B = \{2 + 1/n, n \geq 1\}$$

$$\rightarrow 2 < 2 + 1/n < 3$$

$$\rightarrow B \text{ is bounded: for all } x \in B, 2 \leq x \leq 3$$

$$\rightarrow \sup B = \max B = 3 \text{ (at } n = 1; 2 + 1/1 = 3 \in B)$$

We have to prove: $\inf B = 2$

$$\rightarrow \text{For all } x \in B, x \geq 2$$

$$\rightarrow \text{For all } \varepsilon > 0, \text{ there exists } x \in B \text{ such that } 2 \leq x \leq 2 + \varepsilon$$

$$\rightarrow x = 2 + 1/n < \varepsilon$$

$$\rightarrow n > 1/\varepsilon \text{ (true: } \mathbb{R} \text{ is Archimedean, } n = \lfloor 1/\varepsilon \rfloor + 1)$$

7th session : 15/10/2024

▼ 9. Topology of the line $\mathbb{R}$

Definition: A subset A of $\mathbb{R}$ is said to be an **open set** if for all x in A , there exists an open interval I in A such that:

x belongs to this open interval I .

Examples : Consider the following subsets of $\mathbb{R}$:

$A_1 = (0,1)$ is an open set

$A_2 = (0,1) \cup (10,15)$ is an open set

$A_3 = [0,1) \cup (10,15)$ is not an open set because for $0 \in A_3$, there's no open interval $I \subset A_3$ containing 0. However, for

$x = 0.5 \in (0,1)$, there is an open interval. An open subset of $\mathbb{R}$ is the union of open intervals.

Properties

1. The union of open sets is an open set.
2. The finite intersection of open sets is an open set.
3. The empty set $\emptyset$ and $\mathbb{R}$ are open sets.

Example: The intersection of $(-1/n, 1/n)$ for $n \geq 1$ gives us a set which is not an open set:

$$\bigcap (-1/n, 1/n) = \{0\}$$

Definition: A subset B of $\mathbb{R}$ is said to be a closed set of $\mathbb{R}$ if its complement B^c is an open set of $\mathbb{R}$.

Examples: $B = [1,100] \cup [101,150]$

$$B^c = (-\infty, 1) \cup (100, 101) \cup (150, +\infty)$$

$$\mathbb{N}^c = (-\infty, 0) \cup \bigcup (n, n+1)$$

Neighborhood

Definition: Let x be an element of $\mathbb{R}$. A set A is said to be a neighborhood of x if there exists an open interval I contained in A and containing x .

Examples: $(0,1)$ is a neighborhood of 0.5 because $I = (0,1)$ and $0.5 \in I$. Any open set is a neighborhood of all its elements. $A = [0,1)$ is a neighborhood of all its points, except 0 .

Properties:

1. An open set is a neighborhood of all its elements.
2. The union of neighborhoods of an element is still a neighborhood.
3. The finite intersection of neighborhoods of an element is still a neighborhood.



Theorem

A set is open if and only if it is a neighborhood of all its elements.

▼ Proof:

A is a neighborhood of all its elements:

$$\exists \varepsilon > 0, A \cap (x - \varepsilon, x + \varepsilon) = \{x\}.$$

8th session : 21/10/2024f

14:04 Mon, 21 Oct 35%

< Lecture

2) Hausdorff space (separated space)

$\mathbb{R}$ is an Hausdorff space because

$\forall x, y \in \mathbb{R} : x \neq y \Rightarrow \exists$ a neighborhood for each such that $V_x \cap V_y = \emptyset$

Indeed, let $\varepsilon = |y - x| > 0$

and consider $V_x =]x - \frac{\varepsilon}{2}, x + \frac{\varepsilon}{2}[$, $V_y =]y - \frac{\varepsilon}{2}, y + \frac{\varepsilon}{2}[$

such that: $V_x \cap V_y = \emptyset$

$V_x =]x - \frac{\varepsilon}{2}, x + \frac{\varepsilon}{2}[$

$V_y =]y - \frac{\varepsilon}{2}, y + \frac{\varepsilon}{2}[$

$\varepsilon = |y - x|$

must not intersect

corollary (important)

A is open $\Leftrightarrow$ A is neighborhood of all its elements

proof

$\Rightarrow$ we suppose that A is open so $\forall x \in A, \exists I_x$ open interval

< Lecture

proof

$\Rightarrow$ we suppose that A is open so $\forall x \in A, \exists I_x$ open interval

such that $x \in I_x \subset A$

thus A is a neighborhood of all its elements

$\Leftarrow$

suppose that A is a neighborhood of all its elements:

$$\forall x \in A, \exists I_x^{\text{open int}} : x \in I_x \subset A$$

$$\Rightarrow \{x\} \subset I_x \subset A$$

$$\Rightarrow A = \bigcup_{x \in A} \{x\} \subset \bigcup_{x \in A} I_x \subset \bigcup_{x \in A} A = A$$

$\downarrow$
 A

نقطة التراكم Accumulation point of a set

Def: let A be a subset of $\mathbb{R}$. an element $b \in \mathbb{R}$

is said to be an accumulation point of A iff:

$$\forall \varepsilon > 0,]b - \varepsilon, b + \varepsilon[\setminus \{b\} \cap A \neq \emptyset$$

i.e for all neighborhoods of b , there exist a common element between the neighborhood, and the subset A different than b .

$$A =]0, 1] \quad \text{---} \quad]_{A} \quad \text{---} \quad]_{\frac{1}{5}}[$$

< **Lecture**

$$A = \begin{bmatrix} 0 & 1 \\ 5 & 1 \end{bmatrix}$$

$\forall b: b > 1$, b can't be an accumulation point

$$b < 0, \quad b \text{ ————— } A$$
$$\text{let } x \in A: \forall \varepsilon > 0: [x - \varepsilon, x + \varepsilon] \setminus \{x\} \cap A \neq \emptyset$$

$$\begin{bmatrix} \vdots \\ x_{n-2} \\ x_{n-1} \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \end{bmatrix}$$

0 is also an accumulation point of A

$$\forall \varepsilon > 0,]0 - \varepsilon, 0 + \varepsilon[\setminus \{0\} \cap A =]0, \varepsilon[$$

$$A' = [0, 1]$$

Def: the set of accumulation points of A is denoted by A' (نقطة التجمع)

Example $A =]5, 17] \cup \{4\}$

$$A' = [5, 17]$$

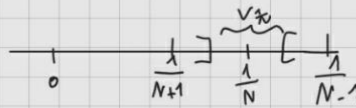
Example

$$A = \left\{ \frac{1}{n+1}, n \geq 0 \right\}$$

prove: $A' = \{0\}$

$$-1 \quad 1 \quad] \quad \overbrace{!}^{V_{\text{K}}} \quad [\quad 1$$

< Lecture



$$A = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{N}, \dots \right\}$$

$\forall b \in A, b$ can't be an accumulation point

indeed: let $b = \frac{1}{N}$, and consider $\varepsilon = \frac{1}{N(N+1)}$. $V_{\frac{1}{N}} = \left] \frac{1}{N} - \frac{\varepsilon}{2}, \frac{1}{N} + \frac{\varepsilon}{2} \right]$

$$\Rightarrow V_{\frac{1}{N}} - \left\{ \frac{1}{N} \right\} \cap A = \emptyset$$

$$\forall \varepsilon > 0, \exists \delta \in \mathbb{R} \cap \{0\} \cap A \neq \emptyset$$

$$\Leftrightarrow \exists \delta \in \mathbb{R} \cap A \neq \emptyset$$

$$\forall \varepsilon > 0$$

$$? \exists n : \varepsilon > \frac{1}{n+1}$$

$$n > \frac{1}{\varepsilon} - 1 \text{ ym } n \text{ exist}$$

$\mathbb{R}$ archimedean

$$n = \left\lceil \frac{1}{\varepsilon} - 1 \right\rceil + 1$$

Isolated points of a set on $\mathbb{R}$

Def: let A be a subset of $\mathbb{R}$. A point $b \in A$

< Lecture



$\mathbb{R}$ archimedean

$$n = \left\lfloor \frac{1}{\varepsilon} \right\rfloor + 1$$

Isolated points of a set on $\mathbb{R}$

Def: let A be a subset of $\mathbb{R}$. A point $b \in A$

$$\exists \varepsilon > 0, \exists b - \varepsilon, b + \varepsilon \cap A = \{b\}$$

$$\exists b - \varepsilon, b + \varepsilon \cap \{b\} \cap A = \emptyset$$

Definition (not programmed)

Let A be a subset of $\mathbb{R}$. we said closure

of A , denoted by $\overline{A}$, the set: $A' \cup \{\text{isolated points}\}$

closed set = contains all
acc points